



Exploring the Impact of Active Transport Amenities on Lifestyle-Related Health Risks: a PLS-SEM Analysis of Diabetes Incidence

Qijie He^{1,2}, Yuanyuan (Anna) Wang³, Zhuo Chen^{2*}

¹School of Transportation, Southeast University; ²Department of Civil and Environmental Engineering, Monash University;

³Department of Public Health and Preventive Medicine, Monash University



Background

1. Diabetes Prevalence

- In 2021, **529 million** people of all ages worldwide have diabetes^[1].
- By 2050, more than **1.31 billion** people are projected to have diabetes^[1].

2. Type 2 Diabetes Risk Factors

- Type 2 diabetes cases made up **96.0%** of all diabetes cases^[1].
- High BMI** was the leading risk factor for type 2 diabetes globally, followed by dietary risks, **environmental risks**, occupational risks, tobacco use, **low physical activity**, and alcohol use^[1].

3. Transportation Environment and Physical Health

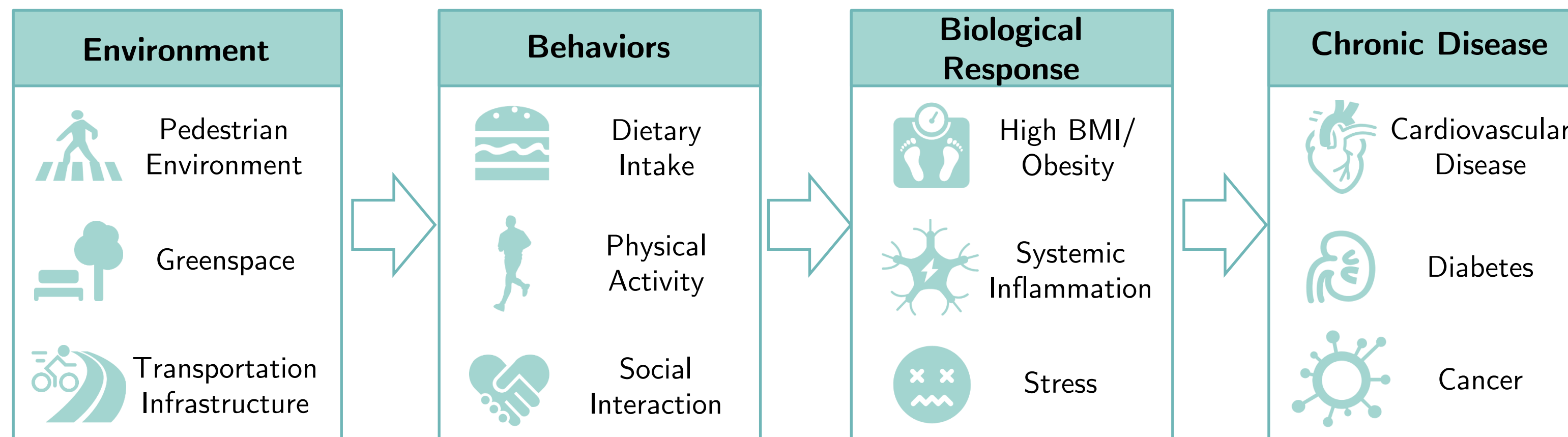


Figure 1. Causal relationship between built environment and chronic disease^[2]

How active transport amenities affect Diabetes incidence?
How to measure the effect?

Study Area

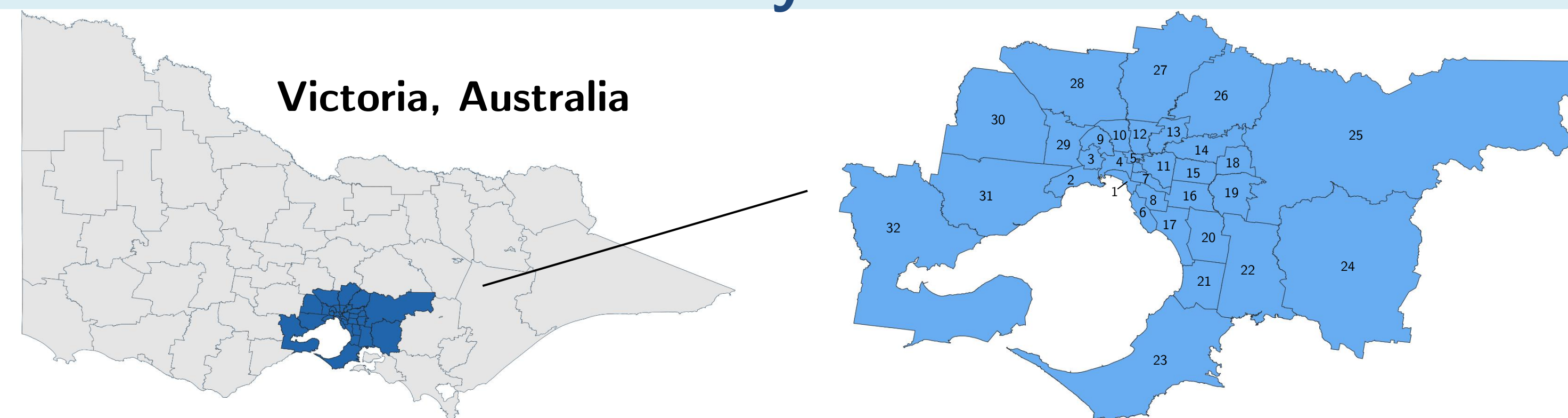


Figure 2. Study area in the state of Victoria, Australia

The data is aggregated across **32 Local Government Areas (LGAs)** in Greater Melbourne, Geelong, and other key regional centers.

Hypotheses

1. Direct Effects

- H1:** Active Transport Amenities positively affect Active Transport Usage.
- H2:** Active Transport Amenities negatively affect Physical Health.
- H3:** Active Transport Usage negatively affects Health Risks.
- H4:** Active Transport Usage negatively affects Physical Health.
- H5:** Health Risks positively affect Physical Health.

2. Indirect/Mediating Effects

- H6:** The effect of Active Transport Amenities on Physical Health is mediated by Active Transport Usage.
- H7:** The effect of Active Transport Amenities on Physical Health is sequentially mediated by Active Transport Usage and Health Risks in succession.

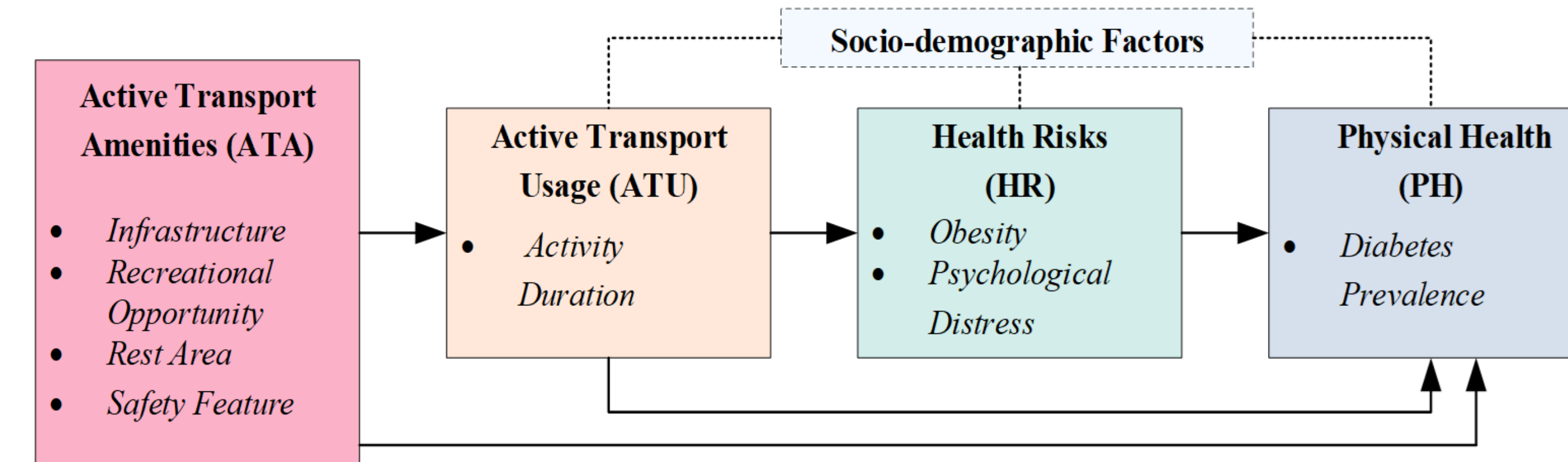


Figure 3. Hypothesized pathways among active transport amenities, usage, health risks, and physical health

Method

Partial Least Squares Structural Equation Modeling (PLS-SEM)

The measurement model:

$$x = \Lambda_x \xi + \delta(1)$$
$$y = \Lambda_y \eta + \epsilon(2)$$

The structural model:

$$\eta = B\eta + \Gamma\xi + \zeta(3)$$

PLS-SEM maximizes the explained variance in the dependent variables to optimize predictive accuracy, does not require distributional assumptions, and is more suitable for small sample sizes and non-normal data.

Results

1. Measurement Model

- All constructs demonstrated acceptable reliability and validity, see Table 1.

Table 1 Measurement Model Evaluation Metrics			
	α	CR	AVE
Active Transport Amenities (ATA)	0.832	0.882	0.612
Active Transport Usage (ATU)	0.857	0.914	0.782
Health Risks (HR)	0.809	0.909	0.834

Note: Cronbach's Alpha (α) > 0.7; Composite Reliability (CR) > 0.7; Average Variance Extracted (AVE) $\geq$ 0.5

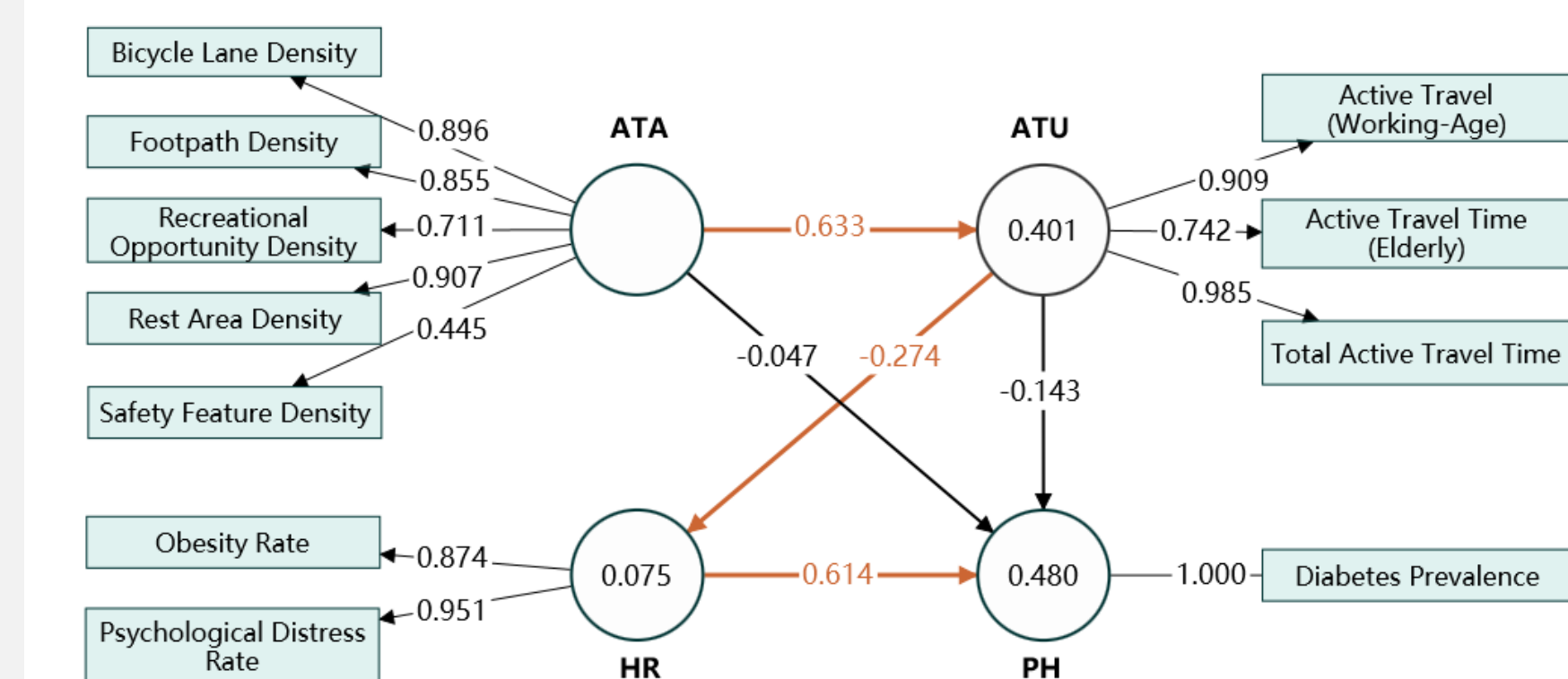


Figure 4(a). Structural model 1: without control variable

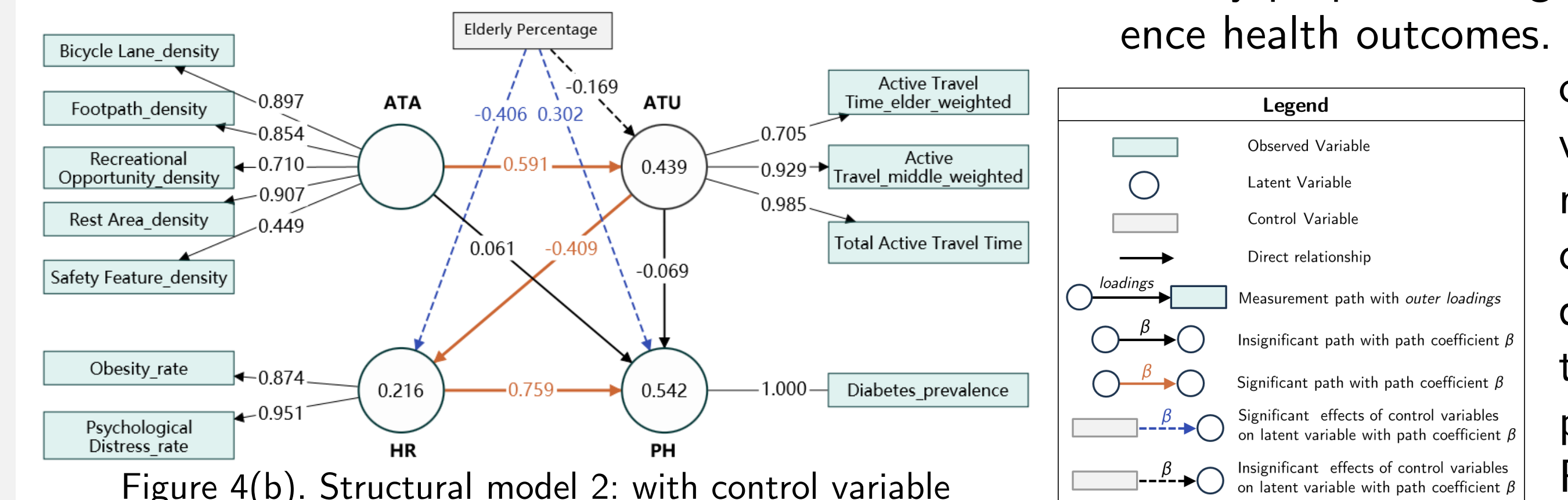


Figure 4(b). Structural model 2: with control variable

2. Structural Model

- Active transport amenities (ATA) positively affect active transport usage (ATU), which in turn reduces health risks (HR) and contributes to better physical health (PH), see Figure 4(a).
- Elderly proportion significantly influence health outcomes. The inclusion of this control variable slightly modified path coefficients but did not change the overall pattern, see Figure 4(b).

Conclusions

Active transport amenities, such as bicycle lanes, footpaths, and recreational areas, have no direct effect but a mediating effect on diabetes incidence, suggesting that improvements in the active transportation infrastructures can support better health outcomes.

References

- [1] Kanyin Liane Ong et al. "Global, regional, and national burden of diabetes from 1990 to 2021, with projections of prevalence to 2050: a systematic analysis for the Global Burden of Disease Study 2021." In: The Lancet, Volume 402, Issue 10397, 2023, p203-234
- [2] Lawrence D. Frank, Nicole Iroz-Elardo, Kara E. MacLeod, Andy Hong. "Pathways from built environment to health: A conceptual framework linking behavior and exposure-based impacts" In: Journal of Transport & Health, Volume 12, 2019, p319-335